PS 2

1 $G_1(s) = \dfrac{2s+11}{s^6+8s^5+17s^4+16s^3+31s^2+8s+15}$

Char eqn.

$a(s) = s^6+8s^5+17s^4+16s^3+31s^2+8s+15 = 0$

- coeffs are non-zero
- all have positive sign (all coefficients must share the same sign)

	column 1	column 2	column 3	column 4
s^6	1	17	31	15
s^5	8/8=1	16/8=2	8/8=1	
s^4	15/15=1	30/15=2	15/15=1	
s^3				
s^2				
s^1				
s^0				
s				

s^4: $\dfrac{1(17)-1(2)}{1} = 15$ (col 2) | $\dfrac{1(31)-1(1)}{1} = 30$ (col 2)

$\dfrac{1(15)-1(0)}{1} = 15$ (col 3)

s^3: $\dfrac{1(2)-1(2)}{1} = 0$ (col 1) | $\dfrac{1(1)-1(1)}{1} = 0$ (col 2)

have to replace row of 0's w/ deriv of auxiliary polynomial

$A_2(s) = s^4+2s^2+1$ col 1 | col 2

$\dfrac{dA_1}{ds} = 4s^3+4s \rightarrow 4 \quad 4 \rightarrow 1 \mid 1$

s^2: $\dfrac{1(2)-1(1)}{1} = 1$ (col 1) | $\dfrac{1(1)-1(0)}{1} = 1$ (col 2)

s^1: $\dfrac{1(1)-1(1)}{1} = 0$

$A_2(s) = s^2+1 \qquad \dfrac{dA_2}{ds} = 2s$

s^0: $\dfrac{2(1)-1(0)}{2} = 1$

First column all +ive → 0 sign changes → 0 poles in right half plane → marginally stable

$$G_2(s) = \frac{s^2+78}{s^5+2s^4+10s^3+40s^2+40s+78}$$

The coeffs in the denominator are of mixed signs → unstable.

$$2.\ P(s) = \frac{6}{s^3+6s^2+11s+6} \qquad C(s) = (K_p + K_I/s) = \frac{K_p s + K_I}{s}$$

$$u(t) = K_p e(t) + K_I \int_0^t e(\tau)\,d\tau$$

Closed loop TF $T(s) = \frac{Y(s)}{R(s)}$

$$T(s) = \frac{C(s)P(s)}{1+C(s)P(s)}$$

$$\therefore C(s)P(s) = \frac{K_p s + K_I}{s} \cdot \frac{6}{s^3+6s^2+11s+6}$$

$$= \frac{6(K_p s + K_I)}{s^4+6s^3+11s^2+6s}$$

Char eqn

$$1 + \frac{6K_p s + 6K_I}{s^4+6s^3+11s^2+6s} = 0$$

$$\frac{s^4+6s^3+11s^2+6s}{s^4+6s^3+11s^2+6s} + \frac{6K_p s + 6K_I}{s^4+6s^3+11s^2+6s} = s^4+6s^3+11s^2+(6+6K_p)s+6K_I = 0$$

- coeffs are non-zero
- all share same sign

Hilroy

s^4	1	11	$6K_I$
s^3	$6/6 = 1$	$(6+6K_p)/6 = (1+K_p)$	0
s^2	$10-K_p$	$6K_I$	
s^1	$\frac{10+11K_p-K_p^2-6K_I}{10-K_p}$		
s			
s^0	$6K_I$		

s^2: $\frac{(1\cdot 11)-(1\cdot(1+1K_p))}{1} = 10-K_p \quad (10>1)$

$\frac{(1\cdot 6K_I)-1\cdot 0)}{1} = 6K_I$

s^1: $\frac{(10-K_p)\cdot(1+1K_p)-(1\cdot 6K_I)}{10-K_p} = \frac{10+9K_p-K_p^2-6K_I}{10-K_p}$

s^0: $\frac{(By\ coeff)\cdot 6K_I-(10-K_p)\cdot 0}{By\ coeff} = 6K_I$

Stable: All first column must be +ve

For $s^2 \rightarrow K_p < 10$

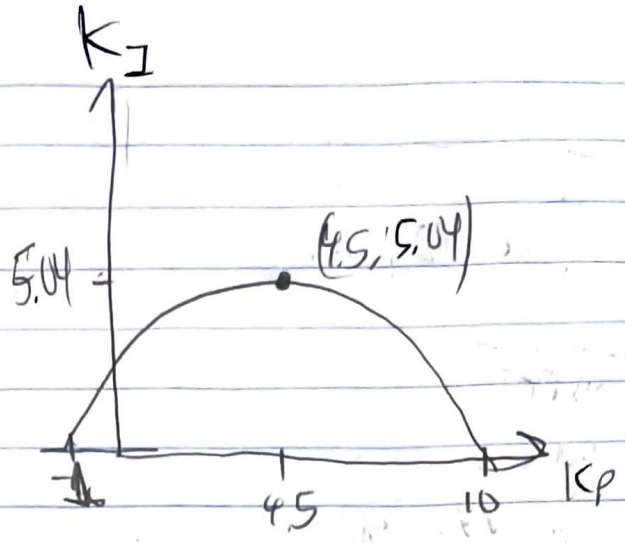
For $s^0 \rightarrow K_I > 0$

For $s^1 \rightarrow 10+9K_p-K_p^2-6K_I > 0$

$$K_I < \frac{10+9K_p-K_p^2}{6}$$

$-(K_p-10)(K_p+1) > 0 \Rightarrow -1 < K_p < 10$

$$\begin{cases} -1 < K_p < 10 \\ K_I > 0 \\ K_I < \frac{-K_p^2+9K_p+10}{6} \end{cases}$$



$K_I = 5.04$ Aom prev formula

3) $J = 600,000\ kg/m^2$ $B = 20000\ N ms$

$J\ddot{\theta} + B\dot{\theta} = T_c$

a) $Js^2\theta(s) + Bs\theta(s) = T_c(s)$

$\theta(s)(Js^2 + Bs) = T_c(s)$

$G(s) = \frac{\theta(s)}{T_c(s)}$

$\theta(s) = \frac{1}{Js^2+Bs} = \frac{1}{s(Js+B)}$

$G(s) = \frac{1}{s(600000s + 20000)}$

b) $T_c = k(\theta_i - \theta)$

$Js^2\theta(s) + Bs\theta(s) = K(\theta_i(s) - \theta(s))$

$\theta(s)(Js^2 + Bs + K) = K\theta_i(s)$

$T(s) = \frac{\theta(s)}{\theta_i(s)} = \frac{K}{Js^2+Bs+K}$

$T(s) = \frac{K}{600,000s^2 + 20000s + K}$

Hilroy

c) $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

Using $s^2 + \frac{B}{J}s + \frac{k}{J} = 0$

$\omega_n^2 = \frac{k}{J} = \frac{k}{600{,}000}$

$2\xi\omega_n = \frac{B}{J} = \frac{20000}{600000} = \frac{1}{30}$

$PO = 100 \cdot e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \leq 10$

$\ln(0.1) = -\frac{\pi\xi}{\sqrt{1-\xi^2}} \Rightarrow \frac{\pi\xi}{\sqrt{1-\xi^2}} = 2.3026$

$\xi = \frac{2.3026}{\sqrt{\pi^2 + 2.3026^2}} = 0.5912$

From $2\xi\omega_n = \frac{1}{30}$

$\omega_n = \frac{1}{60\xi} = \frac{1}{60(0.5912)} \approx 0.0281$

$\omega_n^2 = \frac{k}{J}$

$k = J \cdot \omega_n^2 = 600{,}000 \cdot (0.0281)^2 = 476.9\ Nm$

$k \leq 476.9\ Nm/rad$

d) $t_r = \frac{1.8}{\omega_n}$

$\frac{1.8}{\omega_n} < 80 \Rightarrow \omega_n > \frac{1.8}{80} = 0.0225$

$\omega_n^2 = \frac{k}{600{,}000}$

$\frac{k}{600{,}000} > (0.0225^2) = 0.0005065$

$k > (600{,}000)(0.0005065)$

$k > 303.75\ Nm/rad$

4) $P(s) = \frac{1}{s(s+2)}$ $T_r = 1$ NO 5%

$C(s) = K$

$$T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)}$$

$$T(s) = \frac{\frac{K}{s(s+2)}}{1 + \frac{K}{s(s+2)}} = \frac{K}{s(s+2)+K} = \frac{K}{s^2+2s+K}$$

Char eqn $s^2 + 2s + K = 0$

$s^2 + 2s + K = 0$ $\quad s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$\omega_n^2 = K \Rightarrow \omega_n = \sqrt{K}$

$2\xi\omega_n = 2 \Rightarrow \xi\omega_n = 1 \Rightarrow \zeta = \frac{1}{\sqrt{K}}$

$PO = 100e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}} = 5\%$

$$\xi = \frac{-\ln(0.05)}{\sqrt{\pi^2 + (\ln(0.05))^2}} = 0.690$$

$\sqrt{K} = \frac{1}{\xi} = \frac{1}{0.69} \simeq 1.449$

$K = 1.449^2 = 2.10$

$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$ $\quad \omega_n\sqrt{1-\xi^2} = \sqrt{K}\sqrt{1-\frac{1}{K}} = \sqrt{K-1}$

$T_p = \frac{\pi}{\sqrt{K-1}}$

$K = 2.10$

$T_p = \frac{\pi}{\sqrt{2.1-1}} = \frac{\pi}{\sqrt{1.10}} = 2.995$

Hilroy

$$1 = \frac{\pi}{\sqrt{K-1}} \Rightarrow \sqrt{K-1} = \pi \Rightarrow K-1 = \pi^2$$

$$K = \pi^2 + 1 = 10.87$$

$$\zeta = \frac{1}{\sqrt{10.87}} = 0.303$$

$$PO = 100 e^{-\frac{\pi(0.303)}{\sqrt{1-0.303^2}}} = 36.8\%$$

Not possible to achieve both.

$K = 2.10 \Rightarrow P.O = 5\% \Rightarrow T_r = 3s$

$K = 10.87 \Rightarrow T_r = 1s \Rightarrow P.O = 36.8\%$

K tuning only offers 1 deg of freedom.